

EC5.203 - Communication Theory
Mid Exam

March 02, 2026

- Number of questions: 4 ; Total points: 20; Time Limit: 90 minutes.
- Use of a calculator is permitted.
- This is a closed-book exam.
- Clearly write your assumptions or use of any properties at each step.
- Even if the final answer is correct, only *partial marks* will be given if the approach or methodology is incorrect or is not presented clearly.
- Even if the final answer is incorrect or incomplete, *partial marks* may be given if the approach or methodology is presented clearly.

1. State whether the following statements are True or False. Also, give your reasons in a few (not more than 2-3) sentences! No step-marks for this question! **(5 points)**
 - (a) Communication is also possible between two nearby points, separated in time (by months and years).
 - (b) Envelope detection can be used to demodulate an SSB + carrier signal.
 - (c) For a superheterodyne receiver, the bandpass filter at the RF stage is needed even though there is a bandpass filter at the IF stage.
 - (d) $\delta(t)$ is a baseband signal.
 - (e) The passband waveforms $a_p(t) = u_c(t) \cos(2\pi f_c t)$ and $b_p(t) = u_s(t) \sin(2\pi f_c t)$ are orthogonal for all $u_c(t)$ and $u_s(t)$.
2. For relations between passband and baseband, prove the following
 - (a) Show that the energy of a passband signal equals that of its complex envelope upto a scaling factor: $\|u_p\|^2 = \frac{1}{2}\|u\|^2$. **(2 points)**
 - (b) Show that the correlation of two passband signals equals that of their complex envelope upto a scaling factor: $\langle u_p, v_p \rangle = \frac{1}{2}\Re\{\langle u, v \rangle\}$. **(2 points)**
 - (c) What is the implication of the above results? **(1 point)**

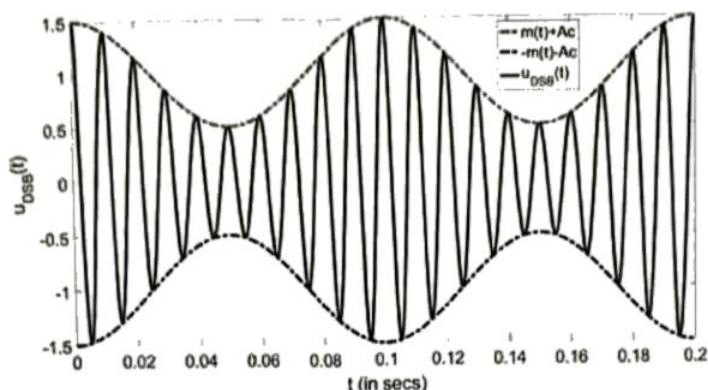


Figure 1: Figure for Question 3.

3. Fig. 1 shows a signal obtained after amplitude modulation by a sinusoidal message. The carrier frequency is difficult to determine from the figure and is not required for answering the following questions

- Find A_c , M_0 , A_m and modulation index. (2 points)
- Find the power in the total modulated signal, information-carrying signal, and power efficiency. (2 points)
- Find the bandwidth of the AM signal. (1 point)

4. Answer the following for the superheterodyne receiver

- What was the motivation for the superheterodyne receiver when it was invented? (1 point)
- Explain the principle of the superheterodyne receiver using a proper block diagram? (2 points)
- Consider that there are 100 FM radio stations with bandwidths of 200 KHz. The central frequency of an FM radio is $88 + 0.2 \cdot S$ MHz, where S is the station index and ranges from 0 to 99. If IF is 2 MHz, how many and which radio stations would interfere with the radio station with index 50? Assume $f_{LO} = f_{RF} + f_{IF}$. (2 points)